Learning Objective 6.3: I can describe how HD Radio works and demodulate an HD Radio signal.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). Use the hybrid numbers from the lesson: subcarrier spacing $\Delta f = 1{,}488{,}375/4096$ Hz, sidebands from subcarrier 356 to 546 on each side, ten partitions of 19 subcarriers per sideband, and a guard of $7/128$ of the useful symbol time.

1. **Finding the sidebands.** A hybrid station broadcasts at 98.5 MHz.

   (a) Find the frequency of subcarriers 356, 450, and 546, in kHz from the channel center, to 0.1 kHz.

   $$\Delta f = 1{,}488{,}375/4096 = 363.37 \text{ Hz}$$
   $$356\,\Delta f = 129.4 \text{ kHz}$$
   $$450\,\Delta f = 163.5 \text{ kHz}$$
   $$546\,\Delta f = 198.4 \text{ kHz}$$

   129.4, 163.5, 198.4 kHz

   (b) Between which two radio frequencies, in MHz, should the upper sideband appear on a spectrum display? The lower sideband?

   Upper: $98.5 + 0.1294 = 98.6294$ MHz to $98.5 + 0.1984 = 98.6984$ MHz. Lower: 98.3016 to 98.3706 MHz.

   upper: 98.6294 to 98.6984

   lower: 98.3016 to 98.3706

   (c) Counting the partition that starts at subcarrier 356 as the first, which partition holds subcarrier 450? The reference subcarriers are 356, 375, 394, and every 19th after that, up to 546. Is subcarrier 451 a reference subcarrier?

   Partition p covers subcarriers $356 + 19(p-1)$ to $356 + 19p - 1$. Since $450 - 356 = 94 = 4 \times 19 + 18$, subcarrier 450 is the last subcarrier of the fifth partition, 432 to 450. Since $451 - 356 = 95 = 5 \times 19$, subcarrier 451 is a reference subcarrier, the first subcarrier of the sixth partition.

2. **Choosing an FFT size.** You record the band with an RTL-SDR at 2.4 MS/s and look at it with a QT GUI Frequency Sink.

   (a) Find the bin width and the number of bins one 69,041 Hz sideband spans, for an FFT of 2048 points and of 16384 points.

$$2048 : 2.4\,\text{MHz}/2048 = 1171.9\,\text{Hz}$$
$$69{,}041/1171.9 = 58.9\,\text{bins}$$
$$16384 : 146.5\,\text{Hz},\ 471\,\text{bins}$$

2048	1171.9 Hz, 59
16384	146.5 Hz, 471

   (b) Which FFT size can locate a sideband's edge to within one 363.4 Hz subcarrier? Which one would you use to find the sidebands quickly on a live band, and why? Why does neither show the subcarriers as separate peaks?

   Only 16384: its 146.5 Hz bins are narrower than a subcarrier, while a 2048-point bin is 3.2 subcarriers wide. To find the sidebands quickly, the 2048-point FFT is enough: 59 bins still draw a clear flat shelf, the display updates and averages faster, and the edges are already within 1.2 kHz. Neither shows separate peaks because the subcarriers are orthogonal: their sinc spectra overlap and add to a flat top, with no dip between neighbors.

3. **OFDM timing and data rate.**

(a) Find the useful symbol time T_u, the guard T_g, the whole symbol time, and the symbol rate.

$$T_u = 1/\Delta f = 2.752 \text{ ms}$$
$$T_g = (7/128)\, T_u = 150.5\ \mu\text{s}$$
$$T_s = T_u + T_g = 2.902 \text{ ms}$$
$$R_s = 1/T_s = 344.5 \text{ symbols/s}$$

T_u 2.752 ms

T_g 150.5 μs

R_s 344.5 /s

(b) Each data subcarrier carries one QPSK symbol per OFDM symbol. Find the raw bit rate of the 360 data subcarriers in both sidebands.

$$360 \times 2 \text{ bits} \times 344.53 \text{ /s} = 248 \text{ kbit/s}$$

248 kbit/s

(c) The hybrid mode delivers about 98 kbit/s of program data. What fraction of the raw rate is that, and what do you think the rest carries?

$98/248 = 0.40$, so about 40%. The rest is redundancy: the bits pass through an error-correcting code before the transmitter maps them to subcarriers, and the code lets the receiver correct the errors that noise and fading cause. The reference subcarriers are not in the 360, so they are extra again.

(d) A reflection from a mountain travels 30 km farther than the direct path. Find its delay. Does it fit inside the guard? How many symbols would it span if the sideband were one carrier with a 14.5 μs symbol?

$$\tau = 30 \text{ km}/(3 \times 10^8 \text{ m/s}) = 100\ \mu\text{s}$$

It fits inside the 150.5 μs guard with 50 μs to spare. On one carrier it would span $100/14.5 \approx 7$ symbols.

τ 100 μs

symbols about 7

4. **Sideband power.** An FM station's analog signal is 6 kW effective radiated power.

(a) At -20 dBc, find the total digital power in watts, the power of each sideband in watts and in dBc, and the power of each of the 382 subcarriers in dBc.

$6000 \times 10^{-2} = 60$ W total, 30 W per sideband

$-20 - 10\log_{10} 2 = -23.0$ dBc

$-20 - 10\log_{10} 382 = -45.8$ dBc

total	60 W
sideband	30 W, -23.0 dBc
subcarrier	-45.8 dBc

(b) The station raises its digital power to -10 dBc total. Repeat (a).

600 W total and 300 W per sideband, which is -13.0 dBc each and -35.8 dBc per subcarrier.

600 W, -13.0, -35.8 dBc

(c) The L14 meters read -26.5 dBc for the upper sideband and -23.1 dBc for the lower on another station. Find the total digital power in dBc, and say what the difference between the two tells you.

$$10\log_{10}\left(10^{-2.65} + 10^{-2.31}\right) = -21.5 \text{ dBc}$$

-21.5 dBc

The sidebands differ by 3.4 dB, so the station runs asymmetric sidebands, most likely to protect a neighbor above it.

(d) On a weak station the shelves stand only 10 dB above the noise floor. Does the meter read the sideband high or low, and by about how much? Why does the frequency sink show the shelves about 25 dB under the host's peak when each sideband is -23 dBc?

High: the meter adds the noise in the band to the sideband's power, and noise 10 dB down adds 10%, which is

$$10\log_{10} 1.1 = 0.41 \text{ dB}$$

The frequency sink shows power per bin, not power per band. The host's power is concentrated in the bins near its center, while each sideband spreads its power evenly over 191 bins, so the host's peak bin is higher relative to a sideband bin than the host's total power is relative to the sideband's total.